ANSI/IES LM-51-20(R2023)

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APPROVED METHOD: ELECTRICAL AND PHOTOMETRIC MEASUREMENT OF HIGH INTENSITY DISCHARGE LAMPS

AN AMERICAN NATIONAL STANDARD

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The IES Testing Procedures Committee

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Foreword

This approved method is a revision of *IES LM-51-2013, IES Approved Method for the Electrical and Photometric Measurement of High Intensity Discharge Lamps*; changes have been made to update the information.

1.0 Introduction and Scope

1.1 Introduction

A high intensity discharge (HID) lamp is an electric-discharge lamp in which the light-producing arc is stabilized by wall temperature, and the arc tube has a bulb wall loading more than 3 W/cm^2. HID lamps include mercury vapor, metal halide, and high pressure sodium lamps.

HID lamps are required to be operated in series with a current limiting device. This auxiliary device, commonly called a ballast, limits the arc current to the value for which each lamp type is designed. It also provides the required starting and operating lamp voltages.

For special purposes, it may be desirable to determine the characteristics of lamps when they are operated at other than the standard conditions described in this Approved Method. Where this is done, such results are meaningful only for the conditions under which they were obtained. All such nonstandard operating conditions shall be stated in the test report.

The photometric information usually required is total luminous flux (lumens) and/or luminous intensity (candelas) in one or more directions, and color. For this Approved Method, the determination of these data will be considered *photometric measurements*.

The electrical characteristics usually measured are line voltage, lamp voltage, lamp current, and lamp power. For this Approved Method, the determination of these data will be considered *electrical measurements*.

1.2 Scope

This Approved Method describes the procedures to be followed and the precautions to be observed in obtaining uniform and reproducible measurements of the electrical and photometric characteristics of high intensity discharge (HID) lamps under standard conditions in alternating current (AC) circuits at 60 Hz. The HID lamps covered by this Approved Method include the lamp groups commonly known as mercury vapor, metal halide, and high pressure sodium used in general lighting applications. Low pressure sodium lamps, covered in *IES LM-59-00, IES Approved Method for the Electrical and Photometric Measurements of Low Pressure Sodium Lamps* (withdrawn from publication), are excluded from this procedure.

2.0 Normative References

2.1 ANSI/IES LS-1-20

Nomenclature and Definitions for Illuminating Engineering. New York: Illuminating Engineering Society; 2020. Online: www.ies.org/standards/definitions/. (Accessed 2019 Feb 25).

2.2 ANSI/IES LM-75-19

Approved Method: IES Guide to Goniometer Measurements and Types, and Photometric Coordinate Systems. New York: Illuminating Engineering Society; 2019.

2.3 ANSI/IES LM-78-20

Approved Method: Total Luminous Flux Measurement of Lamps Using an Integrating Sphere Photometer. New York: Illuminating Engineering Society; 2020.

3.0 Nomenclature and Definitions

The units of electrical measurement used in this test method are the volt, the ampere, and the watt. The units of photometric measurement are the lumen and the candela. Color is specified in terms of CIE recommended systems.[1]

The terms used in this document follow the definitions given in ANSI/IES LS-1-20 (see **Normative Reference 2.1**).

3.1 auxiliary equipment

The reactor or ballast and, if required, starting device needed for the proper operation of discharge lamps. The auxiliary equipment is used to limit lamp current and provide proper starting voltage, and is essential because discharge lamps have a negative volt-ampere characteristic.

Note: For lamps with integrated auxiliary equipment (are selfballasted), subsequent references to auxiliary equipment can be ignored.

3.2 seasoning

The process of operating a new lamp for a fixed period. This is sometimes referred to as aging. A lamp is typically seasoned prior to measuring its initial photometric characteristics. It is an accepted industry procedure for obtaining the "initial" rating of light sources.[2]

4.0 Physical and Environmental Test Conditions

4.1 General

Changes in ambient temperature or air movement have little influence on the electrical and photometric characteristics of HID lamps. The discharge tube is encapsulated inside an outer bulb to provide thermal insulation for the inner discharge tube. Thus, the luminous flux remains almost constant over a wide range of ambient temperatures. It is good laboratory practice that the storage and testing of lamps should be undertaken in a relatively clean environment. Because lamps operate at high bulb temperatures, contaminants can be etched into glass bulb surfaces during operation. Therefore, lamps should be cleaned before measurement.

4.2 Temperature

For practical purposes, an ambient temperature of 25 °C ± 5 °C (77 °F ± 9 °F) is recommended. Although temperature is not critical for HID lamps, it is critical that care be given to the requirements of the measurement instrumentation and the temperature coefficient of the detector.

4.3 Airflow

Airflow does not normally affect the performance of HID lamps. However, special test conditions, such as unjacketed lamps operating in open areas, may require consideration of the effect of airflow.

4.4 Vibration

Lamps should not be subjected to excessive vibration or shock during measurement.

5.0 Electrical Test Conditions

5.1 Line Voltage Waveform

While operating the test lamp, the AC power source, shall have a sinusoidal voltage waveform such that the root-mean-square (RMS) summation of the harmonic components does not exceed 3% of the fundamental.

5.2 Voltage or Current Regulation

The RMS voltage or current of the AC power source shall be regulated to within ±0.1%.

5.3 Power Source Impedance

The power source impedance, measured at the point where the reference ballast and lamp are connected, shall not exceed 2.0% of the reference ballast impedance. Variable autotransformers or other voltage transformation devices should have a volt-amp (VA) rating of at least five times the normal lamp wattage to ensure compliance with this requirement.

5.4 Ballast

When the electrical or photometric characteristics of HID lamps are being measured for rating purposes, the lamp shall be operating in conjunction with a reference ballast of either the appropriate rating or of variable impedance corresponding to each lamp type. The general design features and operating characteristics

* NWL Transformers, Bordentown, NJ, www.nwl.com; and Warner Power, Warner, NH, www.warnerpower.com.

Note: The sources for linear reactors given here are known to the authors of this guide as of October 15, 2011. Providing this information does not in any way imply an endorsement of these companies. Linear reactors suitable for use as reference ballasts may be available from other sources.

of reference ballasts for HID lamps are available.[3] The reference ballast values for voltage and impedance[*] are given on the ANSI lamp data sheets. (See ANSI C78.40-2016,[4] ANSI C78.42-2009 (R2016),[5] ANSI C78.43-2017,[6] or ANSI C78.44-2016,[7] as appropriate.) If readings are to be taken on a lamp type for which no ANSI standard exists, the ballast used shall comply with the general requirements of ANSI C82.5-2016.[8] The procedures to be followed and the precautions to be taken in measuring the performance of ballasts for HID lamps are described in ANSI C78.389-2004 (S2018).[9]

It may be desirable to determine the characteristics of lamps when they are operated by other than reference ballasts. Where this is done, such results are meaningful only for the ballast and circuit with which they are obtained. Such results are not directly comparable with data taken on a reference ballast. Non-reference ballasts shall be documented in the test report.

5.5 Circuits

The measurement circuits employed for HID lamps are shown in **Figure 5-1**. The reference ballasts used here are the series-reactor type. A variable AC power source, capable of providing the input voltage specified for the reference circuit is required. The reference ballast input voltage shall meet the requirements of **Sections 5.1**, **5.2**, and **5.3**.

Figure 5-1 shows the method of connecting into the circuit an instrument capable of measuring and displaying wattage, current, and voltage. The potential element of the instrument is connected on the lamp side of the current-measuring element. An auxiliary voltmeter, shown as V2, may be used if monitoring input and lamp voltage simultaneously is desirable. This type of multifunction instrument usually has negligible effect on the circuit and does not require corrections.

Note: If a VAW instrument with multiple channels is used, the second channel may be used to measure input electrical values of the reference ballast. In this case, the S1 switch and respective wiring may be eliminated from the circuit.

6.0 Lamp Test Procedures

6.1 Lamp Orientation

Lamp seasoning, pre-burning, and photometric measurements shall all be done with the lamp in the same orientation. The operating position should be as specified by the manufacturer. For special application or test purposes, orientation other than that specified by the manufacturer or determined by customary usage may be used if noted in the test report. If no orientation is specified, the base-up position shall be used.

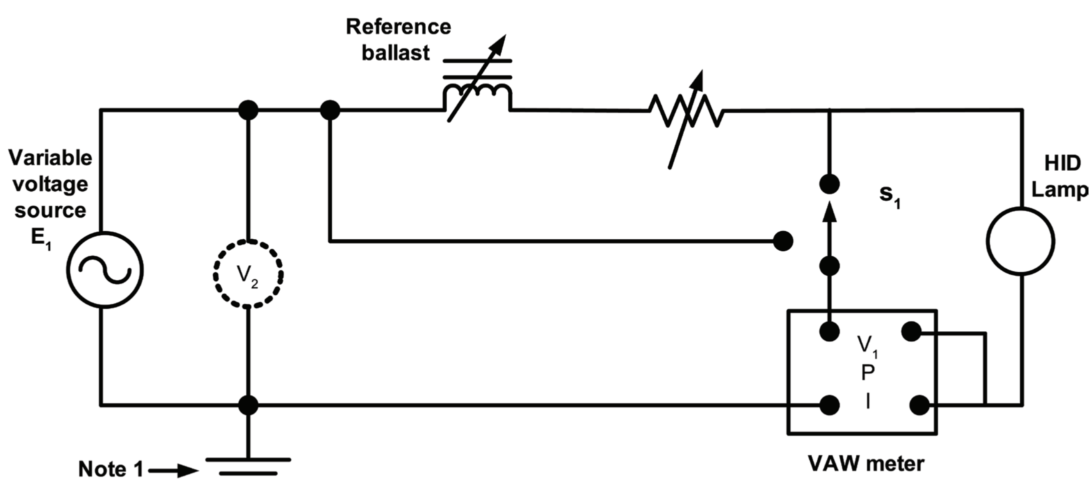

Figure 5-1. The measurement circuit used to test HID lamps includes a variable voltage source (E1) capable of providing the input voltage and current specified for the reference circuit, and a series-reactor type reference ballast. ***Note:*** **This side of the circuit should be connected to ground if a grounded supply system is used.** (© Illuminating Engineering Society)

6.2 Lamp Stabilization

6.2.1 Seasoning. Prior to taking initial measurements, all new high intensity discharge lamps selected for test shall be seasoned per ANSI/IES LM-54-20.[2]

6.2.2 Pre-burning. Before measurements are taken, the lamps should be operated until the electrical characteristics and light output are stable. In most cases, a period of 10 to 30 minutes is sufficient. Metal halide lamps may require considerably longer periods of stabilization time.

However, it is always better to judge stability from periodic checks of light output, lamp voltage (or current), or both, rather than elapsed time. When light output and/or lamp voltage (or current) become stable, the lamp is stable.

A generally accepted method for determining whether an HID lamp is stable includes these steps:

Step 1. Take four measurements of the lamp light output at one-minute intervals (total time = three minutes). This total time is in addition to the recommended pre-burning time.

Step 2. Calculate the percent difference between the maximum measured value and the minimum measured value for the four consecutive measurements.

Step 3. If the value calculated in *Step 2* does not exceed 1% for a mercury vapor lamp, the lamp is considered stable. If the value calculated in *Step 2* does not exceed 2.5% for a high pressure sodium lamp, the lamp is considered stable. If the value calculated in *Step 2* does not exceed 2% for a metal halide lamp, the lamp is considered stable.

Step 4. If the value calculated in *Step 2* exceeds the criteria stated in *Step 3*, the lamp is considered unstable and additional measurements in one-minute intervals should be continued until the last four measurements meet the criteria in *Step 3*.

This method is strongly recommended. Variation in the acceptance criteria or in the number and duration of the measurement intervals provides different levels of lamp stability, either more or less severe. Should a different method or set of criteria be used, it shall be so noted in the test report.

6.3 Electrical Settings

The standard method of photometering an HID lamp shall be with the lamp stabilized and operating on the reference circuit. It shall be read at the rated lamp power. For special purposes, it may be necessary to photometer the lamp with the input volts to the reference circuit set at the rated value or with the lamp current controlled; however, the data obtained from such a measurement is not directly comparable to data obtained using the standard method. The method used shall be indicated in the test report.

6.4 Electrical Instrumentation

6.4.1 Instrument Tolerance. Instrument manuals provide values for the tolerance (accuracy) of the instrument, normally as a percentage of a reading. The instrument tolerance for the measurement of electrical characteristics shall be:

Voltage ±0.5% or less, current ±0.5% or less, and wattage ±0.75% or less.

The reader is referred to NIST 1297 for conversion of instrument tolerance to measurement uncertainty.[10]

6.4.2 Impedance Limitations. Voltage input of the multifunction meter shall have input impedances greater than 1 mega-ohm (MΩ), and the current inputs shall have impedances less than 10 milli-ohms (mΩ).

A test instrument connected in parallel with the lamp shall not draw more than 1% of the rated lamp current. A test instrument connected in series with the lamp shall have an impedance such that the voltage across the instrument does not exceed 2% of the rated lamp voltage.

Caution: Digital test instruments and amplifiers are susceptible to burnout due to starting pulse voltages and high line voltages; therefore, necessary precautions should be taken to have them in the circuit only during lamp measurements and disconnected during lamp ignition.

7.0 Photometric Test Procedures

7.1 Total Luminous Flux Measurements with an Integrating Sphere

7.1.1 Photodetector. The reader is referred to ANSI/IES LM-78-20 (see **Normative Reference 2.3**).

7.1.2 Spectroradiometer. The integrating sphere method gives total luminous flux of the test lamp with one measurement. Appropriate corrections shall be made unless calibration standards agree in spectral power distribution, physical size, and shape with the lamps under test. When the test lamps and the calibrating lamp are not of the same physical size, finish, degree of blackening, and shape, compensation for differences in self-absorption shall be made.

Spectral lumen measurements have many advantages over photodetector lumen measurements. There is no need for spectral mismatch correction due to non-flat sphere paint or the deviation of the photodetector response from V(λ), since every measured wavelength is individually calibrated.[11]

7.2 Measuring Luminous Intensity (Candelas)

Directional light output may be measured with one or more detectors at an appropriate distance from the lamp. The photometer head of the goniophotometer should have a relative spectral responsivity matched to the V(λ) function. The f_1' value (refer to **Annex A**) of the spectral responsivity shall be less than 3%. For absolute measurements of HID sources, a spectral mismatch correction factor should be considered. To determine such a correction factor, the spectral responsivity of the photometer head and the spectral distribution of the source to be measured need to be known. It is important to note that there are photometer heads readily available that have an f_1' value of the spectral responsivity of less than 1.5%.

For determining total luminous flux of a luminaire or lamp using a goniophotometer, the photometer head shall have good cosine response within the angle range where light is incident, with an $f_2(\varepsilon,\phi)$ value (relative deviation from the cosine function) of less than 2% within the acceptance angle range. This good cosine response is required in order to accurately measure the candela values needed to calculate total luminous flux.

7.3 Luminous Intensity Distribution

The distribution of luminous intensity around a lamp may be determined in a photometer like that used for normal luminous intensity measurements, but provided with a means for varying the angles between the detector and the lamp axis. Potential sources of error in addition to those noted in **Section 7.4** that require consideration are:

- Error in the scan angles
- Spectral errors due to the spectral reflectivity of the turning mirror(s)

The detector may be fixed, and the lamp oriented at various angles in planes passing through the axis about which the lamp is moved. Readings are generally taken in 10-degree steps to avoid inaccuracy resulting from sudden changes in distribution.[12]

7.3.1 Intensity Distribution Calculation. The total luminous flux can be calculated from an average intensity distribution curve for a lamp of nominally circular distribution about the beam axis by using lumen constants for annular zones.[13] These zones shall be sufficiently small that negligible error is introduced from variation in intensity. For example, a 10° beam may require 1° zones, while 5° zones may be adequate for a 60° beam. The validity of a flux determination made by this means depends upon the degree to which the intensity distribution represents the true average distribution about the lamp.

7.3.2 Photometer Head (Photodetector). The photometer head of the goniophotometer should have a relative spectral responsivity matched to the V(λ) function. The f_1' value (refer to **Annex A**) of the spectral responsivity shall be less than 3%. For absolute measurements on HID sources, a spectral mismatch correction factor should be considered. To determine such a correction factor, the spectral responsivity of the photometer head and the spectral distribution of the source to be measured need to be known. It is important to note that there are photometer heads readily available that have an f_1' value of the spectral responsivity of less than 1.5%.

For determining total luminous flux of a luminaire or lamp using a goniophotometer, the photometer head shall have good cosine response within the angle range where light is incident, with an $f_2(\varepsilon,\phi)$ value (relative deviation from the cosine function) of less than 2% within the acceptance angle range. This good cosine response is required to accurately measure the candela values needed to calculate total luminous flux.

7.3.3 Spectroradiometer. Spectral lumen measurements have many advantages over photodetector lumen measurements. There is no need for spectral mismatch correction due to non-flat sphere paint or the deviation of the photodetector response from V(λ), since every measured wavelength is individually calibrated.[11]

7.4 Color Measurements

The measurement of color is specified in terms of the chromaticity coordinates, and color rendering indices as defined by CIE.[1,14]

8.0 Test Report

The test report shall list all significant data for each lamp tested, together with performance data. The report shall also list all pertinent data concerning conditions of testing, type of equipment, and lamp information. Typical items reported are:

- Test date and testing agency
- Manufacturer's name and lamp designation
- Type of test
- Number of lamps tested
- Rated electrical values for the lamp tested
- Number of hours operated and operating cycle
- Ambient temperature
- Lamp orientation during test
- Seasoning and stabilization times
- Type of standard (calibration lamp) used and correction factors, if any
- Photometric method
- Sphere diameter or test distance
- Circuit conditions and fixed parameters
- Measured light output (lumens and/or candelas) and electrical values (volts, amperes, watts) of each lamp, and group averages
- Color data
- Spectral power distribution
- Equipment used
- Statement of uncertainty if appropriate
- Deviation from standard operating procedures

Annex A – Measurement Errors Due to the Deviation in System Response from V(λ)

A.1 Illuminance

Commercially available photometric detectors only approximate the standard V(λ) response due to limitations in the manufacturing process. Generally, the closer the "fit" to V(λ), the more expensive the detector. A numerical term used to express the accuracy of fit is f_1'. However, it is important to be aware that this term provides an indication of the average closeness of the detector response to the ideal curve (i.e., its overall conformance to the standard); f_1' does not provide information about the magnitude of deviation of the response from V(λ) at specific wavelengths. To determine the magnitude of error of a measurement due to detector spectral mismatch, the following information is required:

- The relative spectral response of the detector
- The spectral distribution of the source being measured
- The spectral distribution of the source used to calibrate the detector

The relative spectral response of a detector is often available from the manufacturer. Alternatively, it can be measured at several commercial laboratories, provided that the necessary equipment and the expertise are available.[15] The spectral distributions of the source under test and the calibration source are required to be determined using a spectroradiometer.

A.2 Total Luminous Flux

For the measurement of total luminous flux, the detector is positioned at the exit port of an integrating sphere. Integrated, diffuse radiation from a light source is then incident upon the detector. Since the reflectance of the sphere wall varies with wavelength, the relative response of the detector-sphere combination will differ from the response of the detector alone.

As in the case with illuminance measurements (see **Section A.1**), the inaccuracies introduced into the measurement depend on the difference in spectral power distribution of the source under test from that of the calibration source. For example, if the test source and the calibration source are both incandescent lamps operating at the same correlated color temperature, the error in the calibration is canceled out by the same error in the measurement of the test source. On the other hand, if the calibration source is an incandescent source and the test source is a discharge lamp, the error may be significant, even if they are of the same CCT. As in the case of the detector used directly for illuminance measurements, the magnitude of error can be determined and corrections applied using the same method described in **Section A.1**. However, the effects of the sphere need to be considered.

To the novice who wishes to make a simple light output measurement, applying the analytical procedure just described may seem cumbersome. Also, a spectroradiometer is an expensive, complex instrument that may not be readily available. Hence, in the real world, the recommended procedure is not often used, and assumptions are consequently made when a detector is deployed. Unfortunately, such assumptions can be, and often are, incorrect.

The percent error possible in a measurement for any detector is the maximum percent deviation from the V(λ) curve. Knowledge of the detector's response when not accompanied by knowledge of the spectrum of the source under test can lead to large errors, even for a detector with a small f_1' value. An example is the measurement of light emitting diodes (LEDs), which have all their energy in a narrow frequency band.

In some cases, the unknown errors are not significant, and if the person making the measurement ignores them there are no serious consequences. In other situations, relative readings suffice, and absolute levels are not important. If, however, the objective is to obtain a light output measurement with a known level of accuracy, it is critical that the cited references be consulted and the recommended procedures carried out.

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